MERGED CANONICAL FORM · PHASE B

Section 1
Causal Foundation
Zero-Preserving Skeleton

The second deliverable of Phase B: a fully merged, end-to-end rewrite of Section 1 against Construction Spec v1.0. Introduces the **strict partial order $\prec$** on zero-preserving events, the **SOE/MOE two-scale** causal structure, the **emergent speed limit** $c = \gamma \cdot \ell_0$, the **Open Extension Principle** (restated with $\delta\hat{C}$), and the **Two-Link separation** between causal and correlation links. Resolves the §0.8 forward-reference contract for causal order; restores Theorem 1.1.4 (Master-Zero Equivalence) and §1.7 (Emergent Direction/Lorentz) from RCF_n.

DOCUMENT RCF-SEC1-MERGED-v1.0

PHASE B — Section 1 Merge

SCOPE 9 Subsections · 3 Layers · 1 Quarantined Conjecture

SOURCE SPEC RCF-CONST-SPEC-v1.0, Ch. 5–9

DEF 1.1.1 RESTATED | SOE/MOE TWO-SCALE | THM 1.1.4 RESTORED

§0.8 FWD-REF RESOLVED | TWO-LINK SKELETON | §1.7 LORENTZ RESTORED

Preamble — How to Read This Section

This document is the merged canonical form of Section 1 of the Reconciliation Causal Framework (RCF). It is the second deliverable of Phase B as specified in *RCF Unified Construction Specification v1.0*, and it builds directly on the closed foundation established in *RCF Section 0 — Merged Canonical Form v1.0*. Section 0 produced the kinematic algebra, the GNS representation, the Reconciliation Propagator $\mathbf{R}_t$, the thin physical sub-algebra $\mathbf{A}_{phy}^{thin}$ (§0.6), and the full physical sub-algebra $\mathbf{A}_{phy}$ (§0.7, certified equal to the thin candidate by Theorem 0.6.3). Section 1 now introduces the first post-foundational structure: the strict partial order of causal dependency $\prec$, its reconciliation-depth measure, the two-scale (SOE/MOE) speed limit, the Open Extension Principle, the Two-Link separation between causal and correlation links, and the conditional emergence of Lorentzian structure.

The structure follows the spec's source map (Table 4.1) row-by-row. Each subsection opens with a layer badge identifying its position in the L→Q→C→Q emergence ladder, a one-line source citation, and the epistemic tag inherited from the master manuscripts. Body text is ported verbatim where possible; rewritten passages are flagged inline with a spec chapter reference (e.g. *per Ch. 8*). The principal source for this merged section is **RCF_n.pdf §1.0–1.8** (the Gen 1 master manuscript, which uniquely contains §1.1 Master-Zero Equivalence and §1.7 Emergent Direction/Lorentz Compatibility); this is augmented throughout by the post-amendment compact declaration **Section_1_2.pdf** for the SOE/MOE two-scale causal structure introduced by the Gen 3 amendment.

Dependency contract with Section 0

The spec identifies a dependency loop in the Gen 3 draft: §0.4 (SOE flow) used a primitive *"new event E_{new}"* whose only definition was §1.1.1, and §1.1.1 in turn needed $\mathbf{A}_{phy}$ from §0.6. Section 0 v1.0 broke this loop by (i) replacing E_{new} with the raw algebraic perturbation $\delta\hat{C}$ in §0.4 (per Ch. 8), and (ii) splitting $\mathbf{A}_{phy}$ into a thin candidate (§0.6, defined early by ker($\mathbf{M}$)-compatibility) and a full algebra (§0.7, defined late as the fixed-point of $\mathbf{R}_\infty$). **This merged Section 1 closes the contract**: Def 1.1.1 is restated to require only $\mathbf{A}_{phy}^{thin}$ from §0.6 (no forward reference to §0.7 or §0.8), and §1.4 (Open Extension) is restated to use $\delta\hat{C}$ rather than E_{new}. The loop is replaced by an acyclic chain: §0.4 → §0.5 → §0.6 (thin) → §1.1.1 → §1.4 → §1.5.

Forward-reference contract with §0.8 (now partially resolved)

Section 0 v1.0 left a forward reference from §0.8 (Reconciliation Principle) to two Cubic ingredients: **(i)** the causal order $\prec$, and **(ii)** the correlation kernel K_ω. This merged Section 1 supplies ingredient (i): §1.1 defines $\prec$ as a strict partial order on zero-preserving events, and §1.3 derives its two-scale speed limit. Ingredient (ii) remains a forward reference, to be resolved when Section 2 (Emergent Space) is merged. Until then, the variational target I(S) of §0.8 is parameterized on K_ω as a placeholder; once Section 2 is merged, the target becomes fully grounded and the Reconciliation Principle becomes operational.

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§1.0 Purpose of the Causal Foundation

LAYER L → Q

Source: RCF_n.pdf §1.0 (Gen 1 master manuscript, integrated with Section_1_2.pdf §1.0 SOE/MOE split). Epistemic tag: [Established].

Section 0 established the canonical foundation of the framework: a relational algebra $\mathbf{A}$, a family of primitive constraints $\{\mathbf{C}_\alpha\}$, the generated constraint ideal $\mathbf{I}_C$, kinematic states ω_{kin}, the GNS representation, the Reconciliation Propagator $\mathbf{R}_t$, and the thin physical sub-algebra $\mathbf{A}_{phy}^{thin}$ certified equal to the full $\mathbf{A}_{phy}$ by Theorem 0.6.3. The result of Section 0 is an admissibility criterion: it tells us what it means for a relational state to be physical. Physicality means constraint-compatible relational evaluation. However, admissibility alone is not yet structure. It does not tell us what counts as an operation, what counts as an observable, how admissible structures may be transformed, how one admissible event may depend on another, or how causal order can arise without an external time parameter.

The purpose of the present section is to begin that next layer. Section 1 introduces five post-foundational structures: **(1)** zero-preserving events, defined as elements of $\mathbf{A}_{phy}^{thin}$ that preserve ker($\hat{\mathbf{M}}$); **(2)** reconciliation depth d(E), measured at both the SOE and MOE scales; **(3)** causal dependency $\prec$, a strict partial order derived from constraint closure and reconciliation depth; **(4)** the Open Extension Principle, stating that a physical universe is extendible (not closed) under zero-preserving transformations; and **(5)** the Two-Link Principle, separating causal dependency from relational correlation. None of these structures assume a background spacetime, metric, manifold, field equation, or external clock. They describe how physical admissibility may be preserved, related, and extended within the relational constraint framework.

A central architectural commitment of this section — carried over from the Gen 3 SOE/MOE amendment but now resting on the closed foundation of Section 0 — is the two-scale structure of causality. Within a single open extension, events are causally ordered by their position in the local reconciliation sequence: the raw causal tick $\varepsilon = 1/\gamma$ is the duration of one SOE step. Across multiple open extensions, the accumulated causal depth defines a global partial order $\prec$. The physical sector ker($\hat{\mathbf{M}}$) may contain internal record sub-sectors — decohered branches sharing the same coarse-grained geometry (Layer C). The two scales nest: SOE operates within each MOE epoch, and the MOE descent defines the global causal arrow (thermodynamic irreversibility).

Causal order begins as admissibility dependency, not as background time order. Causal precedence is the dependency of zero-closure.

A reader of the legacy 47-document archive will note that this merged Section 1 makes three structural corrections relative to earlier drafts. **First**, Def 1.1.1 is restated to require only $\mathbf{A}_{phy}^{thin}$ from §0.6 (not the full $\mathbf{A}_{phy}$), implementing the $\delta\hat{C}$-loop fix of Spec Ch. 8. **Second**, the Open Extension Principle (§1.4) is restated to use the raw algebraic perturbation $\delta\hat{C}$ rather than the undefined E_{new} of the Gen 3 draft, consistent with the rewritten §0.4 SOE flow. **Third**, two unique subsections from RCF_n.pdf — §1.1 Master-Zero Equivalence (the bridge from algebraic state physicality to represented master-zero) and §1.7 Emergent Direction and Lorentz Compatibility (the conditional derivation of Lorentzian signature from directional isotropy) — are restored. Both were silently dropped in the Gen 3 compact declarations and are ported here in full.

§1.1 Master-Zero Equivalence and Zero-Preserving Events

LAYER Q

Source: RCF_n.pdf §1.1–1.2 (Master-Zero Equivalence + Def of zero-preserving observable); Section_1_2.pdf §1.1 (SOE/MOE depth). Patch: §1.1.1 RESTATED per Ch. 8 to use $\mathbf{A}_{phy}^{thin}$ from §0.6. Epistemic tag: [Established Theorem / Definition].

Section 0 introduced several equivalent ways of expressing physical admissibility. At the primitive algebraic level, physicality is expressed by the quadratic constraint-vanishing condition $\omega(\mathbf{C}_\alpha{}^\dagger \mathbf{C}_\alpha) = 0$ for all α. At the represented Hilbert-space level, the same condition becomes $\pi_\omega(\mathbf{C}_\alpha)\psi = 0$ for vectors ψ in the represented physical sector. At the master-operator level, the many represented constraints may be packaged into one positive operator $\mathbf{M}_\omega = \Sigma_\alpha\, w_\alpha\, \pi_\omega(\mathbf{C}_\alpha)^\dagger\, \pi_\omega(\mathbf{C}_\alpha)$ with strictly positive weights $w_\alpha > 0$, and the physical sector is then described by the single zero condition $\mathbf{M}_\omega\psi = 0$.

This subsection records the exact equivalence between these formulations and then defines zero-preserving events. It is placed at the beginning of the causal foundation because every later notion — observables, causal events, admissible extensions, and dependency order — relies on knowing what it means to preserve the physical zero sector. Master-zero equivalence is the bridge from constraint satisfaction to observable admissibility: it tells us exactly what must be preserved.

Definition 1.1.1 (Zero-Preserving Event) — RESTATED per Ch. 8

The Gen 3 compact declaration defined an event as an element of $\mathbf{A}_{phy}$ such that $\pi_{kin}(E)\ker(\hat{\mathbf{M}}) \subseteq \ker(\hat{\mathbf{M}})$ and $[E, \hat{\mathbf{P}}_0] = 0$. This is circular when $\mathbf{A}_{phy}$ is defined late (as the fixed-point of $\mathbf{R}_\infty$ in §0.7), because §0.4 (SOE flow) and §1.1.1 mutually depend on each other. The canonical restatement uses only the *thin* candidate $\mathbf{A}_{phy}^{thin}$ from §0.6, defined early by spectral compatibility with $\ker(\hat{\mathbf{M}})$ alone. This is well-defined because reconciliation depth d(E) and $\ker(\hat{\mathbf{M}})$-compatibility do not require the full closure property of §0.7.

DEFINITION 1.1.1 — RESTATED (per Spec Ch. 8)

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Definition 1.1.1 (Zero-Preserving Event) — RESTATED per Ch. 8.

  An element E ∈ A_phy^thin  (defined in §0.6 as

  { A ∈ A : π_kin(A) ker(M^) ⊆ ker(M^) })

  is a ZERO-PRESERVING EVENT if

      [E, P^_0]  =  0

  where P^_0 is the asymptotic projector onto ker(M^) produced

    by R_t (Theorem 0.5.1, Section 0).

  The set of all such events is denoted E_phy.

  Remark.
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    The condition [E, P^_0] = 0 says E respects the spectral
    decomposition induced by F^ on ker(M^). This is strictly
    weaker than requiring E ∈ A_phy^full (defined in §0.7 as
    the fixed-point algebra of R_∞), but Theorem 0.6.3 proves
    the two coincide under the stable-mode assumption. Hence
    no content is lost by the early use of A_phy^thin.

  Patch label:  P1 — Def 1.1.1 uses A_phy^thin (not A_phy).
                Breaks the §0.4→§1.1.1→§0.6 dependency loop.
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Definition 1.1.2 (Reconciliation Depth) — Two-Scale

The Gen 3 SOE/MOE amendment splits the reconciliation depth into two scales, reflecting the two-scale structure of the propagator $\mathbf{R}_t$ defined in §0.4. The SOE (Single Open Extension) depth counts reconciliation steps within one open extension; the MOE (Multiple Open Extensions) depth counts descents across the extension history. The total depth is the weighted sum, where the weight N is the extension count.

DEFINITION 1.1.2 — TWO-SCALE DEPTH (per Section_1_2 §1.1)

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Definition 1.1.2 (Reconciliation Depth) — Two-Scale.

  For any event E ∈ E_phy:

    SOE depth:   d_SOE(E) = number of SOE steps within one
                 extension needed to incorporate E.

    MOE depth:   d_MOE(E) = number of MOE descent steps across
                 the extension history to bring E into ker(M^).

    Total depth: d(E) = d_SOE(E)  +  N · d_MOE(E)

  where N is the extension count (number of open extensions
  accumulated in the history leading to E).

  The elementary causal interval is ε = 1/γ, the duration of
  one SOE step. The SOE rate γ is the rate of the SOE spectral-
  flux flow component of R_t (§0.4).
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Definition 1.1.3 (Primitive Causal Relation)

The primitive causal relation $\prec$ is defined by two conjuncts: a depth inequality and a relational connectivity condition. The depth inequality ensures the events are ordered by reconciliation cost; the connectivity condition ensures they are not merely ordered by depth but are relationally connected — the operation E_2 can act on states that have already been acted upon by E_1. Both conjuncts are necessary: depth alone gives a total preorder, and connectivity alone gives an undirected graph. Together they yield a strict partial order on $\mathbf{E}_{phy}$.

DEFINITION 1.1.3 — PRIMITIVE CAUSAL RELATION

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Definition 1.1.3 (Primitive Causal Relation).

  For events E₁, E₂ ∈ E_phy, we say E₁ CAUSALLY PRECEDES E₂,
  written  E₁ ≺ E₂,  iff:

      (i)   d(E₁) < d(E₂)         (depth inequality)
      (ii)  π_kin(E₁) π_kin(E₂) |ψ⟩ ≠ 0
                               (relational connectivity)

  for some |ψ⟩ ∈ ker(M^).

  The relation ≺ is the directed, asymmetric precursor of the
  Lorentzian causal order. It is the first Cubic-layer ingredient
  (the other being K_ω from §2.1) of the Reconciliation
  Principle's variational target I(S) (§0.8).
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Theorem 1.1.4 (Master-Zero Equivalence) — PORT from RCF_n §1.1

Theorem 1.1.4 is the bridge from algebraic state physicality to represented master-zero. It is the structural theorem about positive sums that justifies packaging the many constraints $\{\mathbf{C}_\alpha\}$ into a single master object $\mathbf{M}_\omega$. The theorem appeared in RCF_n.pdf §1.1 (Gen 1 master manuscript) but was silently omitted from the Gen 3 compact declarations, leaving a gap in the deductive chain from $\omega(\mathbf{C}_\alpha^\dagger \mathbf{C}_\alpha) = 0$ to the kernel characterization of $\mathbf{A}_{phy}$. Restoring it is not optional — without it, the use of $\ker(\mathbf{M})$ in Def 1.1.1 above lacks its algebraic justification.

THEOREM 1.1.4 — MASTER-ZERO EQUIVALENCE (restored)

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Theorem 1.1.4 (Master-Zero Equivalence) — PORT from RCF_n §1.1.

  Let  M^_ω = Σ_α w_α · π_ω(C^_α)† π_ω(C^_α)   with w_α > 0
  be a well-defined positive master constraint. Then for every
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  vector ψ in the relevant domain:

      ⟨ψ, M̂_ω ψ⟩ = 0   ⟺   π_ω(Ĉ_α) ψ = 0  ∀α.

  Equivalently:

      ker(M̂_ω)  =  ⋂_α ker π_ω(Ĉ_α).                  (1.1.4)

Proof.
  (⇒)  Assume ⟨ψ, M̂_ω ψ⟩ = 0. Using the positivity identity,
       0 = Σ_α w_α |π_ω(Ĉ_α) ψ|².
       Every summand is non-negative; w_α > 0 forces each
       summand to vanish:  π_ω(Ĉ_α) ψ = 0  for all α.

  (⇐)  Assume π_ω(Ĉ_α) ψ = 0 for all α. Then every term in
       the sum vanishes, so  M̂_ω ψ = 0.   □

  Remark (Why positive weights cannot be relaxed).
    If some weight were zero, the corresponding constraint would
    not contribute; a vector could satisfy M̂_ω ψ = 0 while
    violating that constraint. If negative weights were allowed,
    different terms could cancel in expectation. The condition
    w_α > 0 is therefore essential.

  Patch label:  P2 — Theorem 1.1.4 restored from RCF_n §1.1
                (was silently dropped in Gen 3 compact declarations).
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Three corollaries of Theorem 1.1.4 are used throughout the rest of this section. **First**, the represented physical sector may be written equivalently as $\ker(\hat{\mathbf{M}}_\omega)$ or as the intersection of the kernels of the individual represented constraints; this is the form used in Def 1.1.1. **Second**, the algebraic state physicality condition $\omega(\mathbf{C}_\alpha^\dagger \mathbf{C}_\alpha) = 0$ implies, via the GNS construction, that the cyclic vector Ω_ω lies in $\ker(\hat{\mathbf{M}}_\omega)$ — the implication runs from algebraic state physicality to represented master-zero. **Third**, the effective classical master functional $\mathfrak{M}(x) = \Sigma_\alpha\, w_\alpha |C_\alpha(x)|^2$ obeys the same equivalence: $\mathfrak{M}(x) = 0 \Longleftrightarrow C_\alpha(x) = 0$ for all α, giving the effective classical physical configuration space $\mathscr{X}_{phy} = \mathfrak{M}^{-1}(0)$.

§1.2 Properties of the Causal Order

LAYER C

Source: RCF_n.pdf §1.3 (strict partial order proof); Section_1_2.pdf §1.2 (two-scale structure + record sub-sector disconnection). Epistemic tag: [Established Theorem].

Def 1.1.3 introduces the relation $\prec$ as a binary predicate on $\mathbf{E}_{phy}$. The first question is whether this relation is well-behaved — specifically, whether it forms a strict partial order. A strict partial order must be irreflexive, transitive, and asymmetric. The Gen 1 master manuscript (RCF_n.pdf §1.3) provides the rigorous proof; the Gen 3 compact declaration asserts the result without proof. This subsection ports the Gen 1 proof in full, then states the two-scale structural theorems that the Gen 3 amendment adds on top.

Theorem 1.2.1 (Causal Partial Order)

The relation $\prec$ on $\mathbf{E}_{phy}$ is irreflexive, transitive, and asymmetric — a strict partial order. The proof uses the constraint-closure characterization: define the closure set $\Gamma(E)$ as the set of primitive zero-preserving structures required for E to be admissible as part of a zero-consistent history. Then $E_1 \prec E_2$ iff $\Gamma(E_1) \subsetneq \Gamma(E_2)$ and $\Gamma(E_1)$ is necessary for $\Gamma(E_2)$. Irreflexivity follows because a set cannot be a proper subset of itself; transitivity follows from the transitivity of proper subset inclusion together with Assumption 1.1 (necessity composition); asymmetry follows because $\Gamma(E_1) \subsetneq \Gamma(E_2)$ and $\Gamma(E_2) \subsetneq \Gamma(E_1)$ cannot both hold.

THEOREM 1.2.1 — CAUSAL PARTIAL ORDER

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Theorem 1.2.1 (Causal Partial Order).

  Under Def 1.1.3 and Assumption 1.1 (necessity composition),
  the relation ≺ on E_phy is a STRICT PARTIAL ORDER:

    (i)   Irreflexivity:   ¬(E ≺ E)
    (ii)  Transitivity:    E₁ ≺ E₂  and  E₂ ≺ E₃  □  E₁ ≺ E₃
    (iii) Asymmetry:       E₁ ≺ E₂  □  ¬(E₂ ≺ E₁)

Assumption 1.1 (Necessity Composition).
  If E₁ ≺ E₂ and E₂ ≺ E₃, then the necessity of Γ(E₁) for
  Γ(E₂) together with the necessity of Γ(E₂) for Γ(E₃) implies
  the necessity of Γ(E₁) for Γ(E₃).

  Remark (Defense of Assumption 1.1).
    Necessity composition is the structural analogue of the
    transitivity of logical entailment: if A entails B and B
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entails C, then A entails C. In the constraint algebra, if the consistency of E_2 strictly requires prior satisfaction of E_1, and the consistency of E_3 strictly requires prior satisfaction of E_2, then E_3 indirectly relies on E_1. This is the minimal extra hypothesis required to complete the transitivity proof.

Proof (sketch).

(i) $\Gamma(E) = \Gamma(E)$, so $\Gamma(E) \subsetneq \Gamma(E)$ cannot hold.

(ii) $\Gamma(E_1) \subsetneq \Gamma(E_2) \subsetneq \Gamma(E_3)$ gives $\Gamma(E_1) \subsetneq \Gamma(E_3)$; necessity composes by Assumption 1.1.

(iii) $\Gamma(E_1) \subsetneq \Gamma(E_2)$ and $\Gamma(E_2) \subsetneq \Gamma(E_1)$ is impossible. □

A strict partial order need not compare every pair of events. Two events $E_i, E_j \in \mathbf{E}_{phy}$ may be *causally incomparable*: neither $E_i \prec E_j$ nor $E_j \prec E_i$ holds. A set of mutually incomparable events is called an *antichain*, and serves as a candidate emergent spatial cross-section — its elements are not causally ordered relative to each other, so they may later be identified as space-like separated. This is the structural origin of the Two-Link Principle (§1.5): causal order alone cannot compare antichain elements, so a separate correlation link is required to define their spatial relationship.

Theorem 1.2.2 (Record Sub-Sector Disconnection) — Two-Scale

The Gen 3 SOE/MOE amendment refines the Gen 1 cross-sector disconnection theorem. In the Gen 1 formulation (RCF_n.pdf §1.2), cross-sector events are causally disconnected because operators from different sectors act on orthogonal subspaces. In the Gen 3 formulation (Section_1_2.pdf §1.2), the physical sector $\ker(\hat{\mathbf{M}})$ may contain internal *record sub-sectors* — decohered branches within the one reconciled Hilbert space. Coherence between distinct record states has been suppressed to zero by MOE descent + dephasing. An observer within one record sub-sector cannot causally access events in another. This is the decohered-branch structure, not a fundamental sector decomposition.

THEOREM 1.2.2 — RECORD SUB-SECTOR DISCONNECTION

Theorem 1.2.2 (Record Sub-Sector Disconnection) – Two-Scale.

Within the physical sector $\ker(\hat{M})$, record sub-sectors (Section 7.1) are causally disconnected:

For $E_i \in$ record sub-sector Σ_i and $E_j \in \Sigma_j$ $(i \neq j)$, neither $E_i < E_j$ nor $E_j < E_i$ holds.

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Mechanism.
  Cross-record matrix elements  P^_i ρ_t P^_j  decay
  exponentially to zero under R_t. The decay is driven by
  (a) MOE descent suppressing superpositions of macroscopically
      distinct configurations, and
  (b) the dephasing channel of R_t erasing residual phase
      coherence between records.

In the asymptotic state, operators from different record
sub-sectors act on orthogonal subspaces, so the relational
connectivity condition (ii) of Def 1.1.3 fails.

Remark.
  Record sub-sectors are the branches of Section 7.3, NOT
  separate reconciliation sectors at the foundational level.
  They share the same underlying physical geometry (Layer C,
  coarse-grained metric). This is decoherence, not sector
  decomposition.
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Theorem 1.2.3 (Two-Scale Causal Structure)

The two-scale structure of the propagator $\mathbf{R}_t$ (§0.4: SOE spectral-flux flow + MOE gradient descent + dephasing) induces a corresponding two-scale structure on causality. SOE steps define the local causal grain ε; MOE descent defines the global causal arrow (thermodynamic irreversibility). The two scales nest: SOE operates within each MOE epoch, and the MOE descent provides the global envelope within which the local causal grain has meaning. This nesting is the structural foundation for the emergent thermodynamic arrow of time (Section 3) and for the irreversibility of measurement (Section 7).

THEOREM 1.2.3 — TWO-SCALE CAUSAL STRUCTURE

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Theorem 1.2.3 (Two-Scale Causal Structure).

  SOE steps define the LOCAL causal grain ε = 1/γ.
  MOE descent defines the GLOBAL causal arrow (thermodynamic
  irreversibility).
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The two scales nest: SOE operates within each MOE epoch.

  SOE epoch  ⊂  MOE descent step  ⊂  global causal arrow

Consequence.
  Local causal order (within an SOE epoch) is reversible in
  principle — the SOE flow is Hamiltonian at leading order.
  Global causal order (across MOE epochs) is irreversible —
  MOE descent is gradient flow on the Bures metric of
  ρ_t, monotonically decreasing the obstruction burden
  B_Δ(ρ_t) = Tr(ρ_t F̂).

This is the structural origin of the arrow of time.
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Theorem 1.2.4 (Maximal Causal Chains)

A maximal causal chain $E_1 \prec E_2 \prec \cdots \prec E_n$ has length $n = d(E_n) - d(E_1)$. This theorem connects the order-theoretic structure of $\prec$ to the metric structure of the reconciliation depth d. It is the foundation for the causal-speed limit (§1.3) and for the emergent proper time formula $d\tau = \alpha(B)\cdot d\sigma$ of Section 3. The chain length is bounded below by 1 (a single step) and above by the causal diameter D (Def 1.3.3).

§1.3 The Reconciliation Speed Limit

LAYER C

Source: RCF_n.pdf §1.6 (Emergent Causal Speed Limit); Section_1_2.pdf §1.3 (SOE-scale finite propagation). Epistemic tag: [Conditional Theorem].

A causal partial order by itself does not impose a finite speed limit on propagation. It merely states dependencies; it does not specify how fast influence can propagate across the relational network. Standard physics requires a finite invariant speed c. In RCF, this speed cannot be postulated as a property of a background spacetime metric — there is no background spacetime at this stage. Instead, it must emerge from the interplay between causal depth and relational distance. This subsection derives an emergent causal speed ceiling c_{RCF} from the primitive structures of the theory: the elementary reconciliation interval $\varepsilon = 1/\gamma$, the bounded relational step length ℓ_0, and the triangle inequality for the correlation distance d_ω (to be formalized in Section 2).

Definition 1.3.1 (Causal Interval)

DEFINITION 1.3.1 — CAUSAL INTERVAL

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Definition 1.3.1 (Causal Interval).

  ε = 1/γ   — the elementary causal interval.

  This is the minimal temporal separation between causally
  distinct events: the duration of one SOE step of R_t.

  γ is the SOE spectral-flux rate (§0.4). It is a primitive
  parameter of the propagator, not derived from a background
  metric.
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Definition 1.3.2 (Relational Distance) — Preview

To measure how far a causal influence has propagated, we need a notion of spatial separation. At this stage, the full correlation geometry has not been developed (it belongs to Section 2), but we can preview the primitive correlation kernel K_ω. For two events $E_i, E_j \in \mathbf{E}_{phy}$ with associated local observables O_i, O_j, the relational distance is $d_\omega(E_i, E_j) = -\ell_0 \log K_\omega(O_i, O_j)$, where $\ell_0 > 0$ is a fundamental length scale. Strong correlation ($K_\omega \approx 1$) means small relational distance; weak correlation means large relational distance. The metric properties of d_ω — including the triangle inequality required by Theorem 1.3.4 below — are formally established in Section 2.

Theorem 1.3.3 (Finite SOE Propagation Speed)

The finite rate γ of the SOE component of $\mathbf{R}_t$ imposes a universal bound on causal propagation. Within the physical sector ker($\mathbf{M}$), causal influence propagates at speed $c \leq \gamma \cdot \ell_0$, where ℓ_0 is the fundamental length scale (the maximum relational distance a single primitive causal link can bridge, per Assumption 1.2 below). Equality $c = \gamma \cdot \ell_0$ holds in the continuum limit when ℓ_0 is identified with the Planck length. This is the reconciliation speed limit — the speed of constraint-inconsistency propagation.

THEOREM 1.3.3 — FINITE SOE PROPAGATION SPEED

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Theorem 1.3.3 (Finite SOE Propagation Speed).

  The maximum rate of SOE flux redistribution across relational
  links is

       c  =  γ · ℓ₀                                    (1.3.3)

  where ℓ₀ is the fundamental length scale (the bounded
  relational step length, identified with ℓ_C by Assumption 1.2).
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Assumption 1.2 (Bounded Relational Step Length).

There exists a maximum relational distance $\ell\_C > 0$ that a single primitive causal link can bridge. For any primitive causal link $E\_a < E\_b$:

$$d\_\omega(E\_a, E\_b) \le \ell\_C.$$

Identification ($\ell\_C \equiv \ell_0$).

ℓ_C and ℓ_0 are structurally linked: ℓ_0 is the minimum resolvable relational distance in the emergent geometry, and a single primitive causal link cannot bridge a distance larger than the scale at which correlations remain non-vanishing. We identify $\ell\_C \equiv \ell_0$, unifying the causal and spatial scales of the framework.

This is the core locality assumption of RCF. Causal influence cannot jump arbitrarily far across the correlation geometry in a single reconciliation step.

Definition 1.3.4 (Causal Diameter)

DEFINITION 1.3.4 — CAUSAL DIAMETER

Definition 1.3.4 (Causal Diameter).

$$D = \sup \{ d(E) - d(E') : E, E' \in E\_phy, \; E' < E \}$$

the maximum reconciliation depth spanning causally connected events within the physical sector. D is finite iff the causal order is locally finite (Def 1.3.5 below).

Theorem 1.3.5 (Emergent Causal Speed Limit) — PORT from RCF_n §1.6

Combining the bounded relational step length (Assumption 1.2) with the elementary reconciliation interval $\varepsilon = 1/\gamma$, the effective causal speed v_C is bounded above by $c_{RCF} = \ell_0/\varepsilon = \gamma \cdot \ell_0$. This matches Theorem 1.3.3 exactly, confirming the SOE-scale derivation. The Gen 1 master manuscript (RCF_n.pdf §1.6) provides the full proof via the triangle inequality for d_ω; we port it here.

THEOREM 1.3.5 — EMERGENT CAUSAL SPEED LIMIT

```
Theorem 1.3.5 (Emergent Causal Speed Limit) — PORT from RCF_n §1.6.

  Let (E_phy, <, R_ω) be an RCF causal-relational structure
  satisfying the bounded relational step assumption (1.2).
  Then any causal propagation from E_i to E_j satisfies:

       v_C(E_i, E_j)  ≤  c_RCF,

  where

       c_RCF  =  ℓ₀ / ε  =  γ · ℓ₀.                            (1.3.5)

Proof.
  Let  E_i = E_0 < E_1 < ⋯ < E_N = E_j  be a causal chain.
  By Assumption 1.2 (using ℓ_C = ℓ₀):

       d_ω(E_k, E_{k+1})  ≤  ℓ₀   for every k.

  By the triangle inequality for d_ω (proven in Section 2):

       d_ω(E_i, E_j)  ≤  Σ_k d_ω(E_k, E_{k+1})  ≤  N · ℓ₀.

  The causal depth of this chain is L_C = N. The minimal causal-
  time cost of N elementary steps is τ_C = N · ε.

  Therefore the effective speed is bounded by:

       v_C(E_i, E_j)  =  d_ω / τ_C  ≤  N · ℓ₀ / (N · ε)
                      =  ℓ₀ / ε  =  c_RCF.   □

  Remark.
    c_RCF is not a postulate of a background metric. It is the
    ratio of two primitive scales: the maximum spatial jump per
    causal step (ℓ₀), and the time cost per causal step (ε).
    In the continuum limit, if the framework recovers standard
```

```
    relativistic physics, one identifies  c ↔ c_RCF.
```

Forward reference resolved (partial)

Theorem 1.3.5 depends on the triangle inequality for $d_\omega(\cdot, \cdot)$, which is a Cubic-layer result established in Section 2.5. This is the second forward reference in the spec's contract: §1.3 (speed limit) → §2.5 (triangle inequality). Unlike the §0.8 → §1.1 forward reference (which this merged Section 1 resolves), this one is a one-way dependency — Section 2 does not depend back on Section 1.3. The proof of Theorem 1.3.5 is therefore *conditional* on the Section 2 result, but no circularity is introduced.

§1.4 Open Extension and Causal Growth

LAYER C

Source: RCF_n.pdf §1.5 (Open Extension Principle); Section_1_2.pdf §1.4 (causal horizon). Patch: §1.4.1 REWRITE per Ch. 8 to use δĈ instead of E_new. Epistemic tag: [Conditional Theorem].

The foundation (Section 0) defines what it means for a state or structure to be physical. Zero-preserving events (§1.1) define what it means to remain physical under transformation. Causal dependency (§1.2–1.3) defines how admissible events may depend on one another. But none of these yet says whether the admissible structure is complete, extendible, or dynamically closed. The Open Extension Principle addresses this gap: a physical universe is not merely a closed solution to the constraint equations, but an *extendible* zero-preserving structure — one that admits further admissible continuations. This allows the framework to speak about growth, continuation, history, and non-finality without treating time as primitive.

Definition 1.4.1 (Open Extension) — REWRITE per Ch. 8

The Gen 3 compact declaration defined an open extension as the incorporation of *"a new event E_{new}"* via SOE spectral-flux flow, requiring (1) ∃E with $E \prec E_{new}$, (2) Master-Zero locally preserved, (3) SOE flux redistributing the new obstruction burden. This is circular in the same way as the original Gen 3 §0.4: E_{new} is a primitive of the definition, but its only formal characterization is Def 1.1.1, which requires $\mathbf{A}_{phy}$ (defined late in §0.7). The canonical rewrite replaces E_{new} with the raw algebraic perturbation $\delta\hat{C}$ — a perturbation to the primitive constraint set $\hat{\mathbf{C}} \subset \mathbf{A}$, defined entirely within the Linear/Quadratic layers. This is consistent with the rewritten §0.4 SOE flow, which already uses $\delta\hat{C}$ as its primitive input.

DEFINITION 1.4.1 — OPEN EXTENSION (rewritten per Ch. 8)

```
Definition 1.4.1 (Open Extension) — REWRITE per Ch. 8.

  An OPEN EXTENSION of the physical sector ker(Mˆ) is the

  incorporation of a raw algebraic perturbation δĈ to the
```

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primitive constraint set Ĉ ⊂ A, via SOE spectral-flux flow,
such that:

  (1)  ∃ E ∈ E_phy  with  E < E_new(δĈ)
       (the perturbation-induced event E_new(δĈ) is
        causally downstream of some existing event)

  (2)  Master-Zero is locally preserved within the extension:
       the perturbed master constraint
         Mˆ(Ĉ + δĈ)  =  Σ_α w_α π_ω(Cˆ_α + δĈ_α)† π_ω(Cˆ_α + δĈ_α)
       continues to admit ker(Mˆ) as a non-trivial physical
       sector (ker(Mˆ(Ĉ + δĈ)) ⊇ ker(Mˆ(Ĉ)) ∩ perturbed domain).

  (3)  SOE flux redistributes the new obstruction burden
       δB_Δ = Tr(ρ_t · δFˆ(δĈ))  along the causal frontier,
       where δFˆ(δĈ) = [δĈ, [δĈ, Fˆ]] + O(δĈ³)  (Eq. 8.2 of §0.4).

The perturbation δĈ is defined entirely within Layer L/Q
(no reference to events, causal order, or correlation).
The event E_new(δĈ) is DERIVED from δĈ via the SOE flow,
not primitive.

Patch label:  P3 — Open Extension uses δĈ (not E_new).
              Consistent with §0.4 SOE flow rewrite.
```

A finite SOE rate γ means extension cannot happen instantaneously. The waiting time for one extension step is the causal interval $\varepsilon = 1/\gamma$. This waiting generates the arrow of time at the MOE scale: each MOE descent step accumulates a finite amount of causal depth, and the accumulation is irreversible (Theorem 1.2.3). The Open Extension Principle therefore connects the static admissibility criterion of Section 0 to the dynamic arrow of time of Section 3, without introducing a primitive time parameter.

Theorem 1.4.2 (Causal Horizon)

THEOREM 1.4.2 — CAUSAL HORIZON

```
Theorem 1.4.2 (Causal Horizon).

  At finite MOE time t, the maximum reconciliation depth
```

```
reachable from the kinematic state is

     d_max(t)  =  ⌊ γ · t ⌋                                  (1.4.2)

Events beyond this depth have not yet been incorporated into
the causal structure. The causal frontier advances at the
SOE propagation speed c = γ · ℓ₀ (Theorem 1.3.3).

Proof (sketch).
  Each SOE step contributes depth 1 and consumes time ε = 1/γ.
  In time t, at most ⌊t/ε⌋ = ⌊γ·t⌋ SOE steps can occur.
  Since d(E) = d_SOE(E) + N·d_MOE(E) ≥ d_SOE(E), the maximum
  reachable depth is bounded by the SOE step count.   □
```

Postulate 1.4.3 (Open Extension Principle) — PORT from RCF_n §1.5

A physical universe is not merely a zero-consistent partial order, but an *extendible* zero-consistent partial order. That is, a universe is a structure $\mathscr{U}_s = (\mathbf{E}_s, \prec_s)$ such that $\mathfrak{M}(\mathscr{U}_s) = 0$ and there exists at least one admissible extension $\mathscr{U}_s \to \mathscr{U}_{s+1}$ with $\mathfrak{M}(\mathscr{U}_{s+1}) = 0$. The extension set $\mathrm{Ext}(\mathscr{U}_s) = \{\mathscr{U}_{s+1} : \mathscr{U}_s \to \mathscr{U}_{s+1} \text{ is admissible}\}$ contains all possible zero-preserving continuations of $\mathscr{U}_s$. An admissible structure is *open* if $\mathrm{Ext}(\mathscr{U}_s) \neq \varnothing$; it is *terminal* if $\mathrm{Ext}(\mathscr{U}_s) = \varnothing$. The Open Extension Principle states that a physical universe is open.

POSTULATE 1.4.3 — OPEN EXTENSION PRINCIPLE

```
Postulate 1.4.3 (Open Extension Principle).

  A physical relational structure may continue only through
  admissible zero-preserving extensions:

     Ext(□_s)  ≠  ∅     for all physical □_s.           (1.4.3)

  Equivalent formulations.
    (i)   Effective classical:  □(□_{s+1}) = 0.
    (ii)  Represented:  Ê · ker(M̂)  ⊆  ker(M̂)  for the
          extension operator Ê.
    (iii) Algebraic:  ω_{s+1}(Ĉ_α† Ĉ_α) = 0  ∀α,
          where ω_{s+1} is the extended state.
```

```
Remark (Extension is not external time).
  The label s in  □_s → □_{s+1}  orders stages of admissible
  growth. It is NOT a physical time coordinate. Physical time
  may later be reconstructed from causal chains, extension
  depth, burden weighting, or clock suppression — but it is
  not assumed here.

Remark (Extension-history equivalence).
  Two extension histories that generate the same causal
  dependency records are physically equivalent. If
  e_a ≮ e_b  and  e_b ≮ e_a,  then  (e_a, e_b) ~_hist (e_b, e_a).
  This prevents the extension order from becoming a physically
  observable preferred frame.
```

§1.5 The Two-Link Principle

LAYER C

Source: RCF_n.pdf §1.4 (Two-Link Principle + independence); Section_1_2.pdf §1.5 (SOE/MOE asymmetry). Epistemic tag: [Established Theorem].

Causal dependency alone is not sufficient to reconstruct physical reality. The framework must also explain how space-like structure, distance, locality, and geometric neighborhoods can emerge. These cannot be obtained merely by saying that one event depends on another. Spatial proximity is not the same thing as causal priority. Two structures may be spatially or relationally close without either one being causally prior to the other. Conversely, one event may be causally upstream of another without being nearby in the eventual emergent geometry. This motivates the Two-Link Principle: the framework requires two structurally distinct kinds of link — causal dependency and relational correlation. Causal links support order and eventual time-like structure; relational-correlation links support proximity and eventual space-like structure. The two may interact, but they must not be collapsed into one primitive relation.

Principle 1.5.1 (Two-Link Principle)

PRINCIPLE 1.5.1 — TWO-LINK PRINCIPLE

```
Principle 1.5.1 (Two-Link Principle).

  The framework contains two structurally distinct kinds of link:
```

CAUSAL LINKS: $E_1 < E_2$

– directed, asymmetric, tied to SOE/MOE reconciliation depth

– supports emergent TIME-LIKE structure

CORRELATION LINKS: $K_\omega(E_1, E_2)$

– symmetric, tied to the GNS inner product

– preserved by SOE spectral flow

– supports emergent SPACE-LIKE structure

Neither implies the other. Formally:

$$E_1 < E_2 \quad \nRightarrow \quad K_\omega(E_1, E_2) \text{ is large.}$$

$$K_\omega(E_1, E_2) \text{ large} \quad \nRightarrow \quad E_1 < E_2.$$

The two-link skeleton is the layered graph

$$G_\omega = (E_{phy}, C, R_\omega)$$

where $C = \{ (E_i, E_j) : E_i < E_j \}$ is the causal edge set,

and R_ω is the correlation-link structure determined by ω.

Theorem 1.5.2 (Causal-Correlation Independence)

The two kinds of link are logically independent. It is possible to have causal dependency without strong correlation: $E_i < E_j$ while $K_\omega(E_i, E_j)$ is small. This would represent a case where E_j depends on E_i, but the two event contexts are not close in the emergent correlation geometry. It is also possible to have strong correlation without causal dependency: $K_\omega(E_i, E_j) \approx 1$ while neither $E_i < E_j$ nor $E_j < E_i$ holds. This would represent a case where two contexts are relationally close or highly correlated, but neither is an admissibility predecessor of the other. This independence is necessary for the framework to support both causal ordering and spatial neighborhoods — and, ultimately, for the Lorentzian signature to emerge (Section 2.5).

THEOREM 1.5.2 — CAUSAL-CORRELATION INDEPENDENCE

Theorem 1.5.2 (Causal-Correlation Independence).

$<$ and K_ω are independent structures on E_{phy}. This forces the Lorentzian signature: causal (temporal) and correlation (spatial) directions are distinct algebraic types.

```
The structural independence is the algebraic precursor of the

Lorentzian signature (-, +, +, +):

     ds²  =  −c² dτ²  +  dℓ_ω²

where  dτ  is the burden-weighted causal-depth element

(Section 3)  and  dℓ_ω  is the correlation-distance element

(Section 2). The formal derivation of the signature is given

in Section 2.5 (Cubic Volumetric Consistency) and the

conditional Lorentz-form proposition is stated in §1.7 below.
```

Bridge to Section 2

Theorem 1.5.2 asserts that the Lorentzian signature is forced by the independence of $\prec$ and K_ω, but the formal proof requires the Cubic Volumetric Consistency theorem (Section 2.5), which establishes the triangle inequality for d_ω and the dimensional closure D = 3. Until Section 2 is merged, the Lorentzian signature remains a *conditional* theorem target. The merged Section 1 supplies the causal half of the two-link skeleton; the merged Section 2 will supply the correlation half. The combination is what enables the Reconciliation Principle of §0.8 to become operational.

§1.6 Multi-Sector and Record Structure

LAYER C

Source: Section_1_2.pdf §1.6 (multi-sector & record structure, quarantined). Epistemic tag: [Structural Conjecture].

The two-scale causal structure of §1.2–1.3 admits a natural question: are there reconciliation sectors *beyond* the single physical sector ker($\hat{\mathbf{M}}$)? The Gen 3 compact declaration raises this question explicitly but quarantines the answer from the deductive stack. The foundational layer (Section 0 + Section 1 §1.1–1.5) commits only to the single sector ker($\hat{\mathbf{M}}$); the multi-sector hypothesis is a structural conjecture whose proof would require dynamical arguments from Sections 5 (gravity) and 8 (cosmology). This subsection records the conjecture in its canonical form, with the quarantine flag preserved.

Conjecture 1.6.1 (Cross-Sector Gravitational Interaction) — QUARANTINED

CONJECTURE 1.6.1 — CROSS-SECTOR GRAVITY (QUARANTINED)

```
Conjecture 1.6.1 (Cross-Sector Gravitational Interaction) — QUARANTINED.

  If multiple reconciliation sectors exist beyond ker(Mˆ), they
```

```
may interact gravitationally through MOE-scale burden coupling:

     T_μν^total  =  ⊕_k  T_μν^(k)

where k indexes the (hypothetical) sectors.

This is DISTINCT from the relational burden channel
(Section 5.1.2, 8.3.1), which is the primary dark-matter
mechanism and operates WITHIN the single physical sector.

STATUS:  Quarantined from the deductive stack. Treated as
speculation, not as established structure. The deductive
stack (Sections 2-9) does not depend on this conjecture.
```

Conjecture 1.6.2 (Record Sub-Sector Formation)

CONJECTURE 1.6.2 — RECORD SUB-SECTOR FORMATION

```
Conjecture 1.6.2 (Record Sub-Sector Formation).

  Record sub-sectors form WITHIN ker(M̂) as MOE descent
  suppresses superpositions of macroscopically distinct
  configurations. These are the branches of Section 7.3, NOT
  separate reconciliation sectors at the foundational level.

  Unlike Conjecture 1.6.1, this is consistent with the
  single-sector commitment of the foundation: there is one
  physical sector ker(M̂), but it may contain multiple
  decohered record sub-sectors. An observer within one record
  sub-sector cannot causally access events in another
  (Theorem 1.2.2).

  STATUS:  Structural Conjecture. The proof requires the
  decoherence threshold universality argument of Section 7.2,
  which is currently a Theorem Target (T-7 in Section 9 audit).
```

§1.7 Emergent Direction and Lorentz Compatibility

LAYER C

Source: RCF_n.pdf §1.7 (unique to Gen 1 master; was silently dropped in Gen 3 compact declarations). Patch: P5 — restored in full. Epistemic tag: [Conditional Proposition / Theorem Target].

Theorem 1.3.5 establishes a finite maximum propagation speed c_{RCF}. However, a finite speed alone is not enough to recover special relativity. An anisotropic medium can have a finite maximum speed without possessing Lorentz symmetry. To recover Lorentzian structure, the framework must reconstruct the notion of *direction* from the emergent correlation geometry, and show that Lorentz compatibility appears when the least-burden speed is direction-independent. This subsection ports the unique Gen 1 derivation of emergent direction and the conditional Lorentz form proposition. It is the structural precursor of the Lorentzian signature derivation in Section 2.5.

Definition 1.7.1 (Distance Profile and Relative Displacement)

In standard geometry, direction is represented by tangent vectors. But in this framework, we do not begin with a manifold. We begin with localisable observables and their correlation distances. For each localisable observable $A \in \mathbf{A}_{loc}$, the *distance profile* is the function $\Phi_A(X) = d_\omega(A, X)$, where X ranges over all other localisable observables. Given two distinguishable observables A and B, the *direction* from A towards B is represented by the profile difference $\Delta_{A\to B}(X) = \Phi_B(X) - \Phi_A(X) = d_\omega(B, X) - d_\omega(A, X)$. A direction is thus not an absolute vector — it is the observable-dependent change in the correlation-distance profile induced by shifting from one correlation class to another.

PROPOSITION 1.7.1 — DISPLACEMENT PROFILE PROPERTIES

```
Proposition 1.7.1 (Properties of Displacement Profiles).

  For all A, B, X ∈ A_loc:

    (i)  Antisymmetry:    Δ_{A→B}(X)  =  −Δ_{B→A}(X)

    (ii) Vanishing on diagonal:  Δ_{A→A}(X)  =  0

  Proof.

    (i)  Δ_{A→B}(X) = d_ω(B,X) − d_ω(A,X)

                    = −(d_ω(A,X) − d_ω(B,X))

                    = −Δ_{B→A}(X).

    (ii) Δ_{A→A}(X) = d_ω(A,X) − d_ω(A,X) = 0.   □
```

Definition 1.7.2 (Direction Equivalence and Emergent Tangent Cone)

Two displacements from the same base point should be considered the same direction if they deform the surrounding distance profile in the same way, up to positive rescaling. The ordered pairs (A, B) and (A, C) (with $B \neq A$, $C \neq A$) determine the same emergent direction at A if there exists $\lambda > 0$ such that $\Delta_{A\to B}(X) \approx \lambda \cdot \Delta_{A\to C}(X)$ for all X. The set of equivalence classes of such ordered pairs is the *emergent tangent cone* $T_A^{em}\mathscr{X}_\omega$ at A. This is the first pre-vectorial notion of directional structure inside the emergent metric space.

Theorem 1.7.3 (Direction Requires Distinguishability)

THEOREM 1.7.3 — DIRECTION REQUIRES DISTINGUISHABILITY

```
Theorem 1.7.3 (Direction Requires Distinguishability).

  If  A ~_ω B  (i.e. d_ω(A, B) = 0, meaning A and B represent
  the same emergent point),  then  Δ_{A→B}(X) = 0  for all X.
  Therefore  A → B  defines no non-trivial spatial direction.

  Proof.
    If  A ~_ω B,  then  d_ω(A, B) = 0.  For any X, the triangle
    inequality for d_ω (proven in Section 2) gives:

       d_ω(A, X)  ≤  d_ω(A, B) + d_ω(B, X)  =  d_ω(B, X).
       d_ω(B, X)  ≤  d_ω(B, A) + d_ω(A, X)  =  d_ω(A, X).

    Therefore  d_ω(A, X) = d_ω(B, X),  so  Δ_{A→B}(X) = 0.   □

  Interpretation:  no distinguishability ▯ no direction.
  Direction only exists between correlation-distinguishable classes.
```

Conditional Proposition 1.7.4 (Lorentz Form under Isotropy)

The emergent propagation ceiling may depend on the direction of the relational displacement. Let $\hat{n}$ denote an emergent direction in $T_A^{em}\mathscr{X}_\omega$. The speed ceiling might be a function $c_\star(\hat{n})$. The propagation ceiling is *direction-independent* (isotropic) if $c_\star(\hat{n}) = c_\star$ for all $\hat{n} \in T_A^{em}\mathscr{X}_\omega$. Directional isotropy means that the least-burden reconciliation speed is the same in all emergent relational directions. This removes preferred spatial axes from the leading-order propagation law. Under directional isotropy and continuum regularity, the effective interval can be written in Lorentz form.

CONDITIONAL PROPOSITION 1.7.4 — LORENTZ FORM UNDER ISOTROPY

```
Conditional Proposition 1.7.4 (Lorentz Form under Isotropy).

  If the emergent propagation ceiling is independent of direction
```

and the continuum limit is regular, then the effective interval can be written in Lorentz form:

$$ds^2 = -c_\star^2\, d\tau^2 + d\ell_\omega^2 \tag{1.7.4}$$

where $c_\star$ is the direction-independent speed ceiling,
$d\tau$ is the burden-weighted causal-depth element (Section 3),
$d\ell_\omega$ is the correlation-distance element (Section 2).

Proof sketch.
Once $c_\star$ is direction-independent, the only remaining first-order invariant compatible with the causal structure is a quadratic interval with one time-like and spatially isotropic part. The sign choice is fixed by the causal ordering: time-like separation must correspond to burden-weighted causal evolution ($d\tau$), while space-like separation corresponds to relational displacement ($d\ell_\omega$).

STATUS: Conditional Proposition. A full proof would require demonstrating that no higher-order or anisotropic terms survive the coarse-graining limit. This is Theorem Target T-4 in the Section 9 audit.

Conditional Proposition 1.7.5 (Emergent Mass Shell)

CONDITIONAL PROPOSITION 1.7.5 — EMERGENT MASS SHELL

Conditional Proposition 1.7.5 (Emergent Mass Shell).

Assume the effective geometry is Lorentz-compatible with invariant interval $ds^2 = -c_\star^2\, d\tau^2 + d\ell_\omega^2$, and assume the excitation carries a conserved burden-weighted four-momentum p_μ. Then the leading-order dispersion relation is:

```
     p_μ p^μ  =  −m_eff² c_⋆²                                  (1.7.5)

where  m_eff  is the effective rest mass, emergent from the

burdened relational excitation.

Remark.

  The burden sector shifts the effective inertial response, so

  m_eff is NOT primitive — it is derived. The full derivation

  of m_eff from burden is given in Section 4.2 (Theorem 4.2.2:

  m ≡ B₀ as identity, with the spectral gap demoted to the

  operator-level measurement of that burden).
```

Status of Lorentz symmetry (theorem-safe language)

It is crucial to maintain theorem-safe status language regarding Lorentz symmetry throughout the framework. **(1)** The speed bound ($v_C \le c_{RCF}$) is a *conditional theorem* — proven from bounded relational step length (Assumption 1.2) and finite reconciliation tick $\varepsilon = 1/\gamma$. **(2)** Directional isotropy ($c_\star(\hat{n}) = c_\star$) is a *regime assumption* — it emerges if the coarse-grained correlation geometry is sufficiently homogeneous. **(3)** Lorentz symmetry itself is an *effective continuum theorem target* — it follows if isotropy and regularity hold. The framework does NOT assume Lorentz symmetry at the foundational algebraic level. It derives it as an effective continuum limit of the relational causal structure.

§1.8 Architectural Summary

The table below summarizes the structural units introduced by this merged Section 1, their layer in the L→Q→C→Q emergence ladder, their best source, and their epistemic status. Every unit is placed at the layer where its ingredients first become available; no unit depends on a layer below it. The two forward-reference contracts (to Section 2 for the triangle inequality of d_ω, and to Section 3 for the burden-weighted causal depth $d\tau$) are documented in the rightmost column. The §0.8 forward-reference contract (causal order $\prec$ as one of two Cubic ingredients of the RP variational target) is now *resolved* by §1.1 of this section.

§	Unit	Layer	Source	Status	Notes / Forward Refs
§1.1.1	Zero-Preserving Event	Q	RCF_n §1.2 + Sec_1_2 §1.1	Established	P1: restated to use A_phy^thin (Ch. 8)

§	Unit	Layer	Source	Status	Notes / Forward Refs
§1.1.2	Reconciliation Depth (two-scale)	Q	Sec_1_2 §1.1	Established	SOE + MOE split
§1.1.3	Primitive Causal Relation ≺	C	RCF_n §1.3 + Sec_1_2 §1.1	Established	Resolves §0.8 fwd-ref (i)
§1.1.4	Master-Zero Equivalence	Q	RCF_n §1.1	Established	P2: restored (was dropped in Gen 3)
§1.2.1	Causal Partial Order	C	RCF_n §1.3	Established	Proof requires Assumption 1.1
§1.2.2	Record Sub-Sector Disconnection	C	Sec_1_2 §1.2	Established	Mechanism: MOE + dephasing
§1.2.3	Two-Scale Causal Structure	C	Sec_1_2 §1.2	Established	SOE grain ⊂ MOE arrow
§1.2.4	Maximal Causal Chains	C	RCF_n §1.3 + Sec_1_2 §1.2	Established	Length n = d(E_n) − d(E_1)
§1.3.3	Finite SOE Propagation Speed	C	Sec_1_2 §1.3 + RCF_n §1.6	Conditional	Requires Assumption 1.2
§1.3.5	Emergent Causal Speed Limit	C	RCF_n §1.6	Conditional	Fwd-ref: §2.5 triangle inequality
§1.4.1	Open Extension (δĈ)	C	RCF_n §1.5 + Ch. 8 rewrite	Conditional	P3: uses δĈ (not E_new)
§1.4.3	Open Extension Principle	C	RCF_n §1.5	Postulate	Universe is open, not closed
§1.5.1	Two-Link Principle	C	RCF_n §1.4 + Sec_1_2 §1.5	Established	≺ vs K_ω
§1.5.2	Causal-Correlation Independence	C	RCF_n §1.4	Established	Forces Lorentzian signature
§1.6.1	Cross-Sector Gravity	C	Sec_1_2 §1.6	Quarantined	Speculation, not in deductive stack
§1.6.2	Record Sub-Sector Formation	C	Sec_1_2 §1.6	Structural Conjecture	Requires §7.2 decoherence threshold
§1.7.1	Distance Profile & Direction	C	RCF_n §1.7	Established	P5: restored (was dropped in Gen 3)
§1.7.3	Direction Requires Distinguishability	C	RCF_n §1.7	Established	Fwd-ref: §2 triangle inequality
§1.7.4	Lorentz Form under Isotropy	C	RCF_n §1.7	Conditional Proposition	Theorem Target T-4

§	Unit	Layer	Source	Status	Notes / Forward Refs
§1.7.5	Emergent Mass Shell	C	RCF_n §1.7	Conditional Proposition	Fwd-ref: §4.2 mass-burden identity

Table 1.8.1 — Architectural summary of Section 1. 20 structural units across 3 layers (L→Q→C). 3 patches implemented (P1, P2, P3, P5); 1 quarantined conjecture (§1.6.1); 2 forward references to Section 2 (triangle inequality, K_ω); 1 forward reference to Section 3 (dτ burden weighting); 1 forward reference to Section 4 (mass-burden identity). The §0.8 forward reference (causal order ≺) is RESOLVED by §1.1.3.

The conceptual sequence of this section is the strict emergence chain: zero sector (from §0.5) → zero-preserving events (§1.1.1, using A_phy^thin from §0.6) → causal dependency ≺ (§1.1.3, §1.2) → two-scale speed limit (§1.3) → open extension (§1.4, using δĈ) → two-link separation (§1.5) → emergent direction & Lorentz compatibility (§1.7, conditional). Each link in this chain depends only on the previous links and on the closed foundation of Section 0. No link depends on a structure introduced later in the chain, and no link depends on Section 2 or beyond (except for the two documented forward references, both of which are one-way).

Section 1 is now CLOSED. Section 2 (Emergent Space) can be merged against this stable causal foundation. Section 2.1 will define the correlation kernel K_ω — the second Cubic ingredient of the §0.8 Reconciliation Principle's variational target I(S). Section 2.5 will prove the Cubic Volumetric Consistency theorem, which establishes the triangle inequality for d_ω required by Theorems 1.3.5 and 1.7.3, and which closes Open Target 1 (metricity). Once Section 2 is merged, the §0.8 forward-reference contract is fully resolved and the Reconciliation Principle becomes operational, completing the Cubic layer of the L→Q→C→Q emergence ladder.